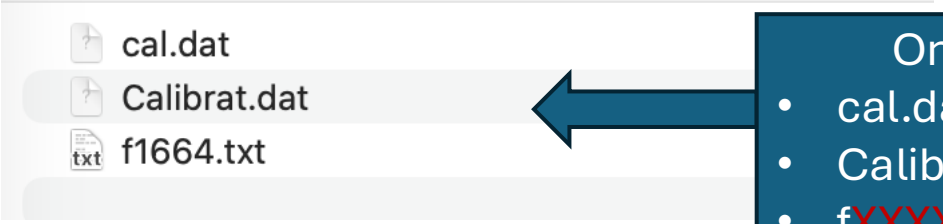
Data conversion – binary to text format (FL30)

Work through example on

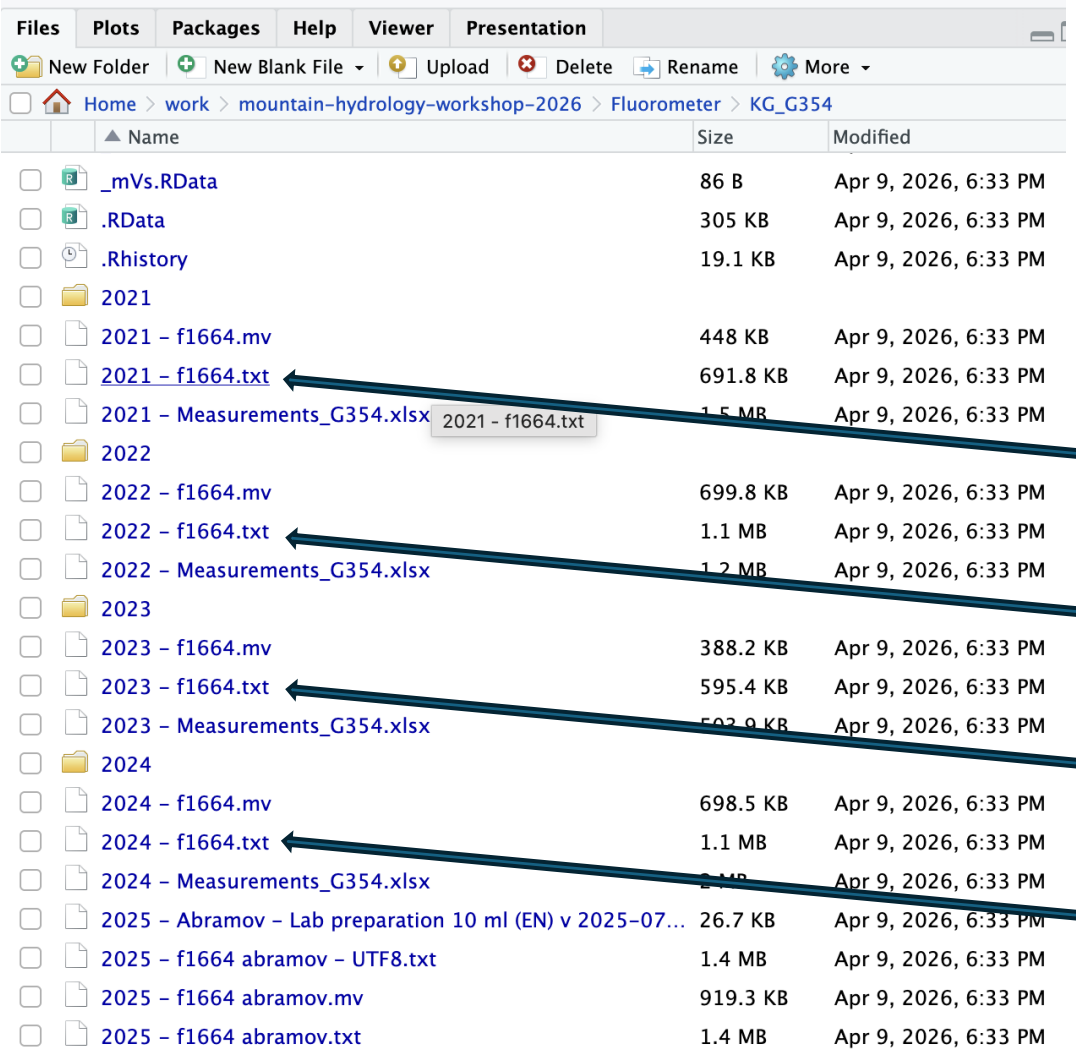
https://renkulab.io/p/eric.pohl/discharge-calculation-dilution-method

In RStudio on renkulab.io (same workflow on your own computer)



On the SD-card of the fluorometer are 3 files

- cal.dat
- Calibrat.dat
- fXXXX.txt (XXXX is the serial number of the logger)



I call each year's file according to the

- Year, e.g. 2022
- Logger number, e.g. f1664

→ "2022 - f1664.txt"

The files need to be converted ...

2021 – f1664.txt

```
A 28:09:21 06:19:32 A 001 37887  001 48137  196 05344  014 29159
A 28:09:21 06:19:34 A 001 37320  001 47541  196 02775  014 29450
A 28:09:21 06:19:36 A 001 39103  001 47350  195 65196  014 29977
A 28:09:21 06:19:38 A 001 37882  001 47723  195 59838  014 31219
A 28:09:21 06:19:40 A 001 37915  001 47821  195 55148  014 31983
A 28:09:21 06:19:42 A 001 37430  001 48206  195 52948  014 32968
A 28:09:21 06:19:44 A 001 37075  001 47512  195 46366  014 33919
A 28:09:21 06:19:46 A 001 37887  001 46920  195 43999  014 34950
A 28:09:21 06:19:48 A 001 37775  001 47327  195 38080  014 36041
A 28:09:21 06:19:50 A 001 37390  001 47699  195 32807  014 37123
A 28:09:21 06:19:52 A 001 37981  001 46662  195 30638  014 38268
```

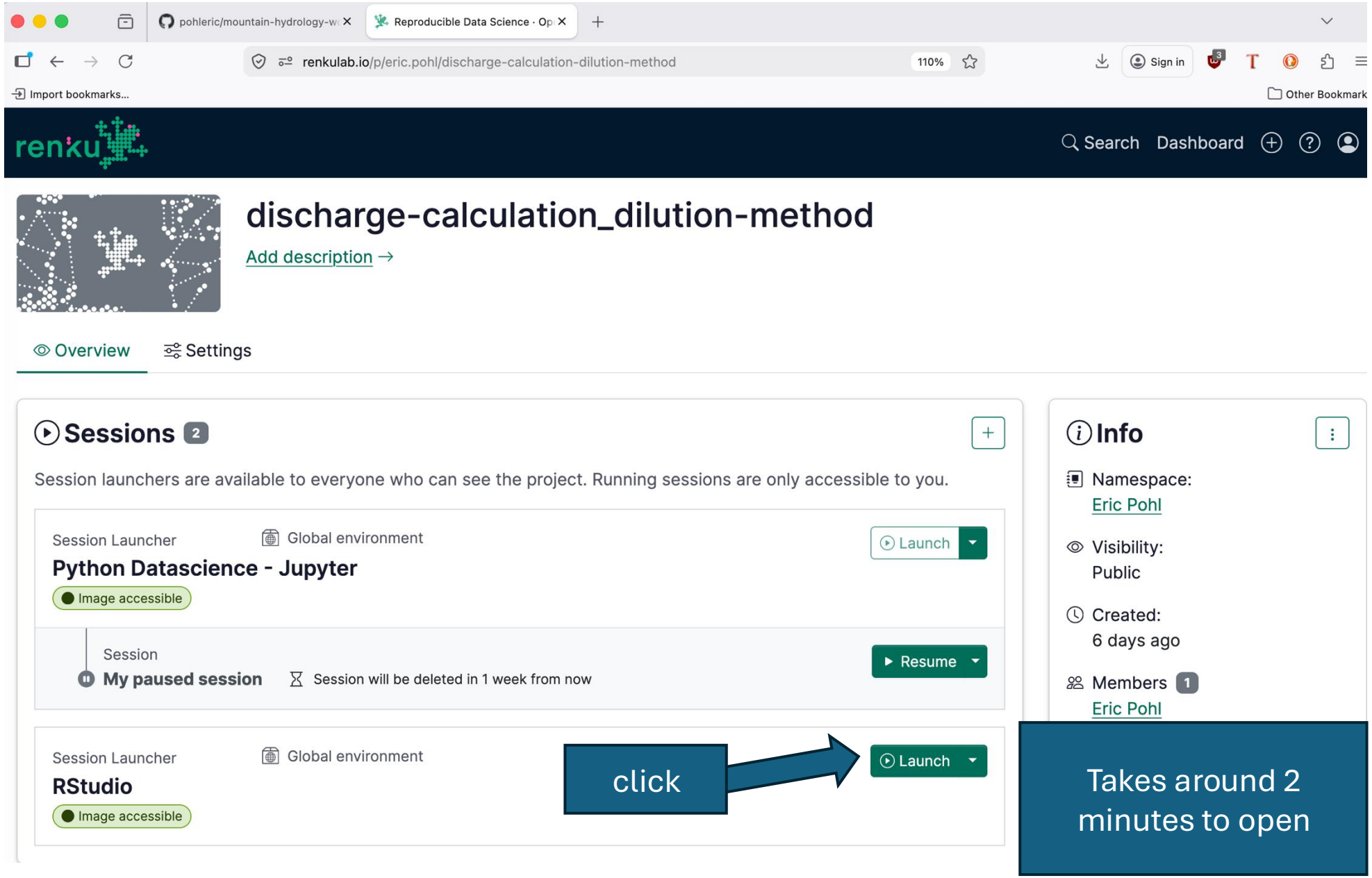
Before conversion (binary format)

2021 – f1664.mv

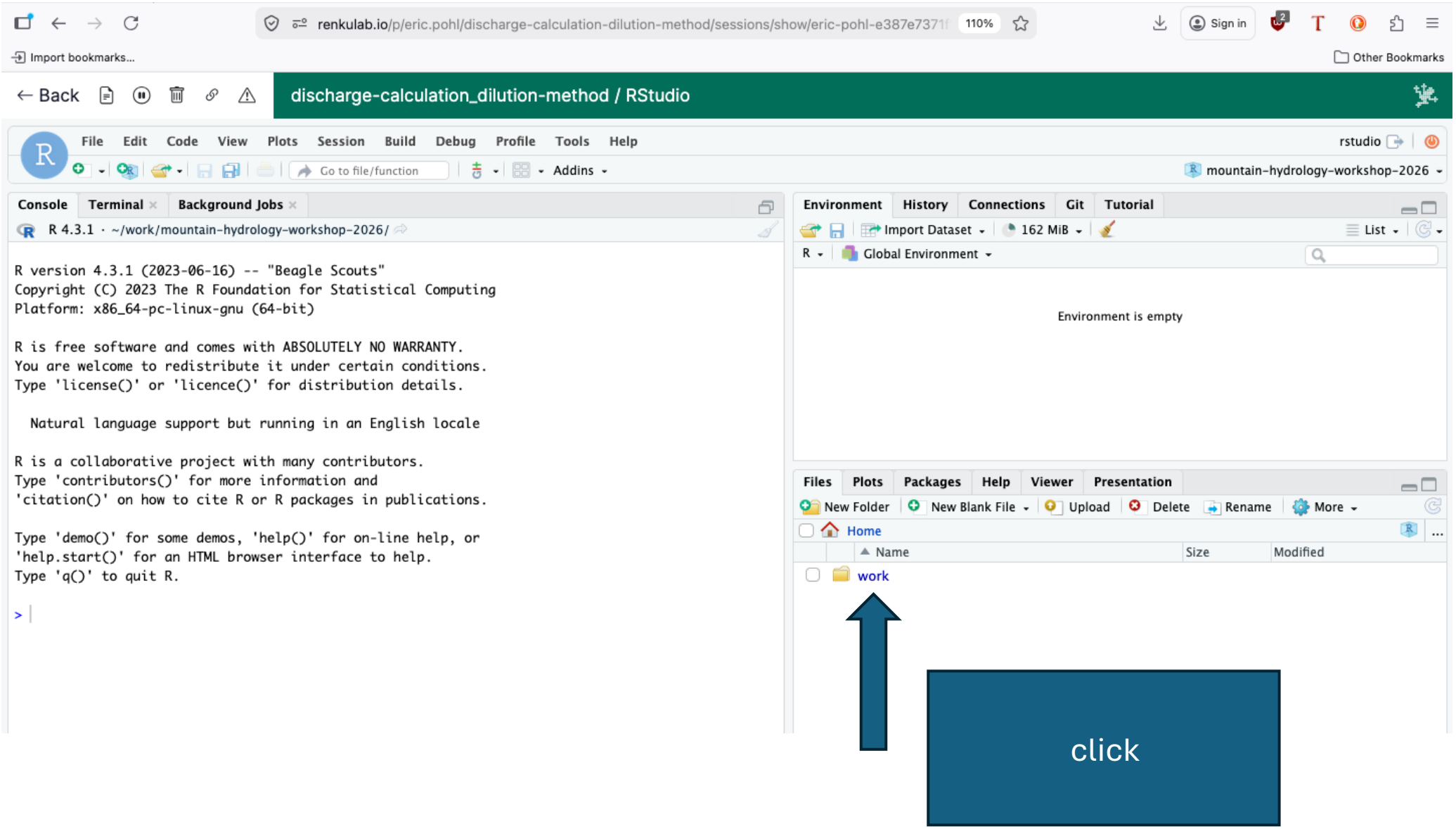
```
date Tracer1 Battery Baseline Temperature
2021-09-28 06:19:32 16.94 11.61 15.41 14.11
2021-09-28 06:19:34 16.85 11.61 15.33 14.11
2021-09-28 06:19:36 16.82 11.61 15.59 14.12
2021-09-28 06:19:38 16.88 11.6 15.41 14.14
2021-09-28 06:19:40 16.89 11.6 15.42 14.15
2021-09-28 06:19:42 16.95 11.6 15.34 14.16
2021-09-28 06:19:44 16.85 11.59 15.29 14.18
2021-09-28 06:19:46 16.76 11.59 15.41 14.19
2021-09-28 06:19:48 16.82 11.58 15.39 14.21
2021-09-28 06:19:50 16.87 11.58 15.34 14.23
```

After conversion

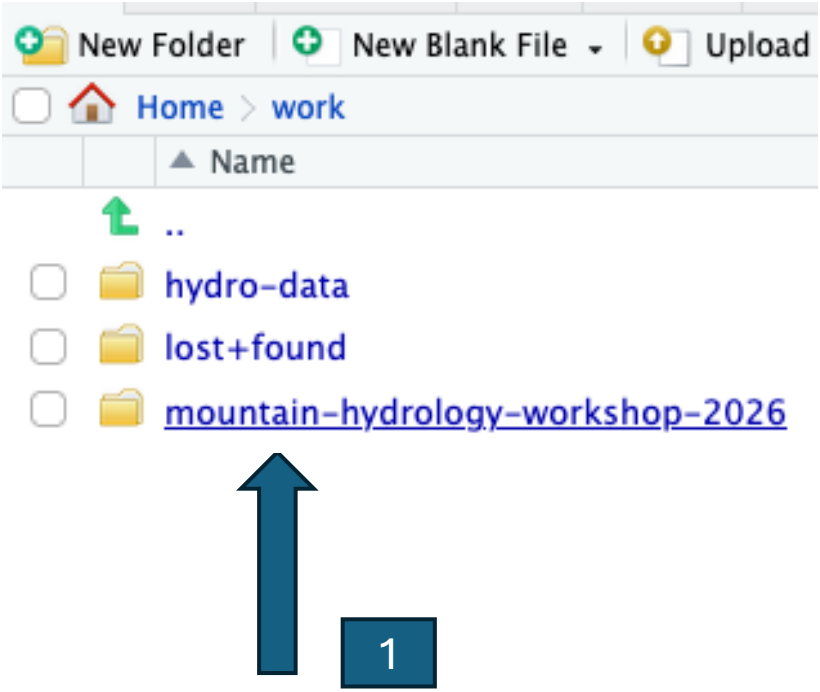
https://renkulab.io/p/eric.pohl/discharge-calculation-dilution-method

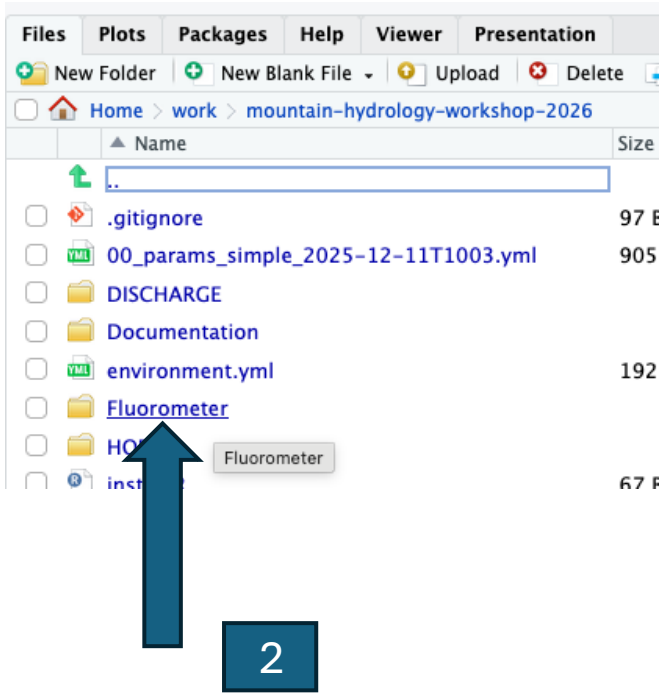


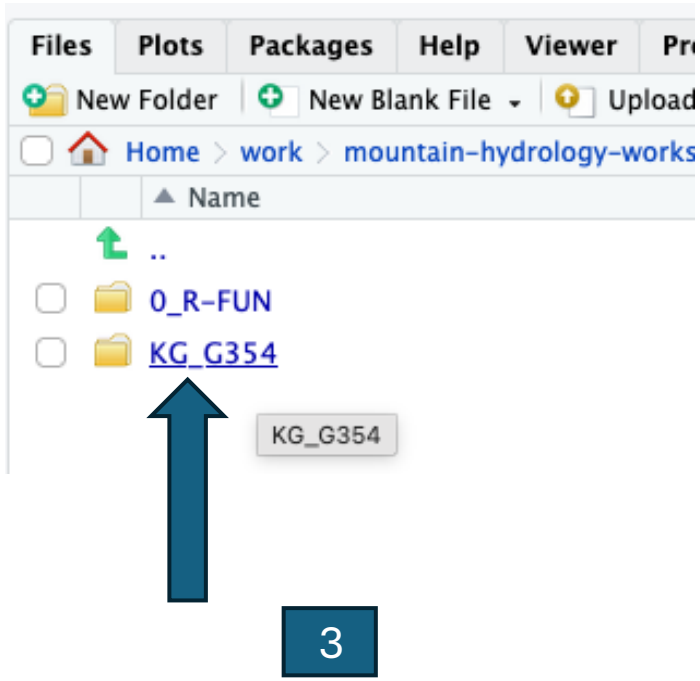
Opening the program. Directory: **work/CA-hydroworkshop/materials/Fluorometer/KG_G354/R/003_Qcalc.R**

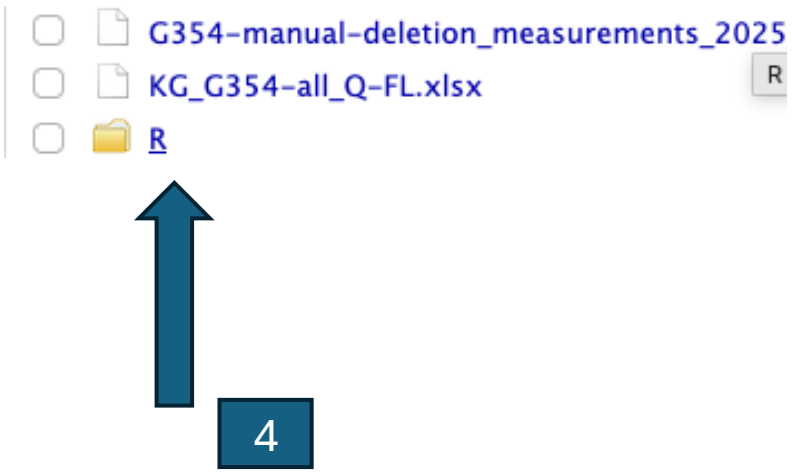


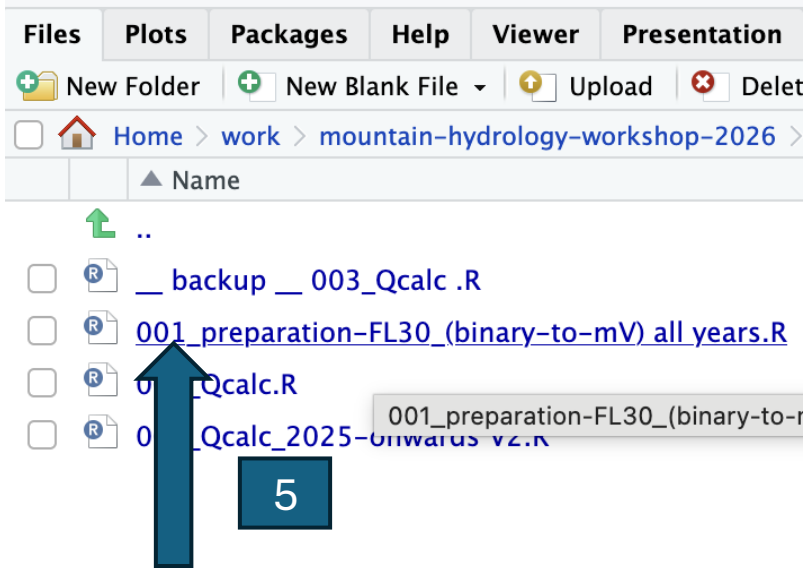
Opening the program. Directory: **work/CA-hydroworkshop/materials/Fluorometer/KG_G354/R/001_prep....R**











Click, click, click ...

This is the conversion “script” (program)

001_preparation-FL30_(binary-to-...

Source on Save Run Source

```r
# - reading FL30 raw data
# - transforming data into readable format with signal in mV

# Arguments
# path = "/Users/pohle/Dropbox/Central_Asia/Fluorometer/data/KG_G354/"
path = "~/work/mountain-hydrology-workshop-2026/Fluorometer/KG_G354/"

setwd(path)
# For Windows it will be something like this:
# path = "C:/Downloads/mountain-hydrology-workshop-2026/Fluorometer/KG_G354/"


# Input files
filename_raw = '2021 - f1664.txt'
# Output files
filename_mV = '2021 - f1664.mv'


# repeat for 2022, 2023, 2024, ...



# ------------ NO NEED TO CHANGE ANYTHING HEREAFTER --------------- #
source("../0_R-FUN/FUN_binary_to_mv.R")
```

This is the conversion “script” (program)

001_preparation-FL30_(binary-to-...

Source on Save | Run | Source

```r
# - reading FL30 raw data
# - transforming data into readable format with signal in mV

# Arguments
# path = "/Users/pohle/Dropbox/Central_Asia/Fluorometer/data/KG_G354/"
path = "~/work/mountain-hydrology-workshop-2026/Fluorometer/KG_G354/"

setwd(path)
# For Windows it will be something like this:
# path = "C:/Downloads/mountain-hydrology-workshop-2026/Fluorometer

# Input files
filename_raw = '2021 - f1664.txt'
# Output files
filename_mV = '2021 - f1664.mv'


# repeat for 2022, 2023, 2024, ...



# ------------ NO NEED TO CHANGE ANYTHING HEREAFTER --------------- #
source("./0_R-FUN/FUN_binary_to_mv.R")
```

The program is designed that you only have to change very few things!

These things are:

- Define the input name
- Define the output name

That’s it. You don’t need to change anything else!

```r
# repeat for 2022, 2023, 2024, ...



# ------------ NO NEED TO CHANGE ANYTHING HEREAFTER --------------- #
source("../0_R-FUN/FUN_binary_to_mv.R")

data_in = bit_to_mv(data_file = paste0(path, filename_raw))
tail(data_in)
head(data_in)

plot(data_in$date,data_in$Tracer1, type='l', xlab='Date', ylab='Signal [mV]')

# write data (mV and normal CSV format)
write.table(aaa,file = paste0(path,filename_mV),quote = F, row.names = F)
```

Scroll down to where it says:
"# ------------ NO NEED TO CHANGE ANYTHING HEREAFTER --------------- #

- Click into this line
- Run line by line [Ctrl]+[Enter]
- You will see graphs produced

- It will write the mV output file

Possible issues

- Prepare the original input file:
  - Check the dates and remove all lines that are not from the measuring period
  - Some loggers have a wrong date (usually maximum a few days off)
  - Start from the last entry (check with your filed notebook when the last measurement was made)
  - Identify the date and time offset
  - Use that information to remove non-relevant data entries

- Corrupt file from the FL30
  - → Error when running the script/program
  - Solution:
    - Read error message
    - It will say where in the original file the error occurs
    - Remove the line
    - Try again
- Switched interval (e.g. from 2 seconds to 30 seconds)
  - → Error when running the script/program
  - Solution:
    - Cut out (delete) the part; this part will look different in the original file